

Let y be an arbitrary element of $A \cap B$, i.e. $y \in A \cap B$.
 By Defn. 2, $y \in A \wedge y \in B$. By Defn. 2, $y \in A$ is True and $y \in B$ is True.
 By Defn. 3 from Section 2.1, $y \in \overline{B}$. By defn. of AND, $y \in B \wedge y \in \overline{B}$. By Defn. 2,
 $y \in B \cap \overline{B}$. By Prob 9(b), $y \in \emptyset$. $\therefore$ By the defn. of empty set, y does not exist.
 $\therefore A \cap B = \emptyset$ $\square$

d) $A \cap B = B \cap A$
Solution: We cannot conclude anything regarding the sets A and B since this is a commutative law of sets and we have verified that this always holds from Prob 11(b). $\square$

e) $A - B = B - A$

Solution: $\therefore A - B \subseteq B - A$ and $B - A \subseteq A - B$.
 By Defn. 4, $x \in A \wedge x \notin B$ if x is an arbitrary element of $A - B$.
 $\therefore$ By Defn. of AND, $x \in A$ and $x \notin B$ both holds.
 By Defn. 3 from Section 2.1, $x \in B - A$. By Defn. 4, $x \in B \wedge x \notin A$. By Defn. of AND, $x \in B$ and $x \notin A$ both holds.
 By Defn. of AND, $x \in A \wedge x \notin A$ is True. By Defn. 5, $x \in A \wedge x \in \overline{A}$. By Defn. 2, $x \in (A \cap \overline{A})$. By Prob 9(b), $x \in \emptyset$. By Defn. of empty set, x does not exist.
 $\therefore A - B = \emptyset$. By Defn. 4, if y is an arbitrary element of $A - B$, then $y \in A \wedge y \notin B$. By Defn. of AND, $y \in A$ and $y \notin B$ both hold. $\therefore A - B = \emptyset$,
 $\therefore$ There does not exist an element which is a member of A and is not a member of B .
 Since an element can either be a member of B or not a member of B ,
 $\therefore$ If x is any arbitrary element of B , it also must be a member of B .
 By Defn. 3 from Section 2.1, $A \subseteq B$.

Similarly, by assuming x is an arbitrary element of $B - A$, we can prove,
 $B \subseteq A$. $\therefore A = B$ $\square$

32) Can you conclude that $A = B$ if A, B and C are sets such that
 a) $A \cup C = B \cup C$? Solution: $A = \{1, 2\}$ $B = \{3, 4\}$ $C = \{1, 2, 3, 4\}$
 $A \cup C = B \cup C = \{1, 2, 3, 4\}$. But $A \neq B$. $\square$